

**Multiple Choice:**

1. The sole purpose of a commutator in a dc generator is to
 - A. increase output voltage
 - B. reduce sparking at brushes
 - C. provide smoother output
 - D. convert the induced ac into dc
2. In a dc motor, the mechanical output power actually comes from
 - A. field system
 - B. air-gap flux
 - C. back emf
 - D. electrical input power
3. The commercial efficiency of a shunt generator is maximum when its variable loss equals _____ loss.
 - A. constant
 - B. stray
 - C. iron
 - D. friction and windage
4. If the field circuit of a loaded shunt motor is suddenly opened
 - A. it will race to almost infinite speed
 - B. it would draw abnormally high armature current
 - C. circuit breaker or fuse will open the circuit before too much damage is done to the motor
 - D. torque developed by the motor would be reduced to zero
5. In dc generators, armature reaction is produced actually by
 - A. its field current
 - B. armature conductors
 - C. field pole winding
 - D. load current in armature
6. If the load current and flux of a dc motor are held constant and voltage applied across its armature is increased by 10 percent, its speed will
 - A. decrease by about 10 percent
 - B. remain unchanged
 - C. increase by about 10 percent
 - D. increase by 20 percent
7. When load is removed, _____ motor will run at the highest speed.
 - A. shunt
 - B. cumulative-compound
 - C. differential compound
 - D. series
8. The basic requirement of a dc armature winding is that it must be
 - A. a closed one
 - B. a lap winding
 - C. a wave winding
 - D. Either B or C
9. The maximum torque by dc motors is limited by
 - A. commutation
 - B. heating
 - C. speed
 - D. armature current
10. In a dc generator, the generated emf is directly proportional to the
 - A. field current
 - B. pole flux
 - C. number of armature parallel paths
 - D. number of dummy coils
11. If the pole flux of a dc motor approached zero, its speed will
 - A. approach zero
 - B. approach infinity
 - C. no change due to corresponding change in back emf
 - D. approach a stable value somewhere between zero and infinity
12. The counter emf of a dc motor
 - A. often exceeds the supply voltage
 - B. aids the applied voltage
 - C. helps in energy conversion
 - D. regulates its armature voltage
13. The speed of a dc motor can be controlled by varying
 - A. its flux per pole
 - B. resistance of armature current
 - C. applied voltage
 - D. All of the above
14. In small dc machines, armature slots are sometimes not made axial but are skewed. Though skewing makes winding a little more difficult, yet it results in
 - A. quieter operation
 - B. slight decrease in losses
 - C. saving of copper
 - D. Both A and B
15. If load on a dc shunt motor is increased, its speed is decreased due primarily to
 - A. increase in its flux
 - B. decrease in back emf
 - C. increase in armature current
 - D. increase in brush drop
16. In a dc motor, unidirectional torque is produced with the help of
 - A. brushes
 - B. commutator
 - C. end-plates
 - D. Both A and B
17. A dc motor can be looked upon as dc generator with the power flow
 - A. reduced
 - B. increased
 - C. reversed
 - D. modified
18. The critical resistance of the dc generator is the resistance of
 - A. armature
 - B. load
 - C. field
 - D. brushes
19. The flux control method using paralleling of field coils when applied to a 4-pole series dc motor can give _____ speeds.
 - A. 2
 - B. 4
 - C. 3
 - D. 6
20. The primary reason for providing compensating windings in a dc generator is to
 - A. compensate for decrease in main flux



- B. neutralize armature mmf
C. neutralize cross-magnetizing flux
D. maintain uniform flux distribution
21. The percentage slip in case of synchronous motor is
- A. 1% C. 100%
B. 0.5% D. zero
22. The main function of interpoles is to minimize _____ between the brushes and the commutator when the dc machine is loaded.
- A. friction
B. sparking
C. current
D. wear and tear
23. A shunt dc motor works on ac mains
- A. Unsatisfactorily
B. Satisfactorily
C. Not at all
D. None of the above
24. A 4-pole generator with 16 coils has a two-layer lap winding. What is the pole pitch?
- A. 32 C. 16
B. 8 D. 4
25. The external characteristic of a shunt generator can be obtained directly from its _____ characteristic.
- A. internal
B. open-circuit
C. load-saturation
D. performance
26. The most economical method of finding no-load losses of a large dc shunt motor is _____ test.
- A. Hopkinson's
B. Swinburne's
C. Retardation
D. Field's
27. The primary reason for making the coil span of a dc armature winding equal to a pole pitch is to
- A. obtain a coil span of 180 degrees (electrical)
B. ensure the addition of emfs of consecutive turns
C. distribute the winding uniformly under different poles
D. obtain a full-pitch winding
28. The shaft torque of a dc motor is less than its armature torque because of _____ losses.
- A. copper
B. mechanical
C. iron
D. rotational
29. If in the case of a certain dc armature, the number of commutator segments is found either one less or more than the number of slots, the armature must be having a simplex _____ winding.
- A. wave C. lap
B. multielement D. frog leg
30. A lap wound DC generator having 250 armature conductor runs at 1200 rpm. If the generator emf is 200 V, then the operating flux of the DC generator is
- A. 0.02 Wb C. 0.08 Wb
B. 0.04 Wb D. 0.06 Wb
31. The normal value of the armature resistance of a dc motor is
- A. 0.005 C. 0.5
B. 10 D. 100
32. In a clockwise-rotating loaded dc generator, brushes have to be shifted
- A. clockwise
B. counterclockwise
C. either A or B
D. neither A nor B
33. The dc series motor should never be switched on at no load because
- A. the field current is zero
B. the machine does not pick up
C. the speed becomes dangerously high
D. it will take too long to accelerate
34. The mechanical power developed by the armature of a dc motor is equal to
- A. armature current multiplied by back emf
B. power input minus losses
C. power output multiplied by efficiency
D. power output plus iron losses
35. The series field of a short-shunt dc generator is excited by _____ currents.
- A. shunt C. armature
B. load D. external
36. What is the rotor input power when rotor copper loss in an induction motor is 600 W and the slip is 3%?
- A. 18 kW C. 200 kW
B. 20 kW D. 25 kW
37. For making coil span equal to a pole pitch in the armature winding of a dc generator, the back pitch of the winding must equal the number of
- A. commutator bars per pole
B. winding elements
C. armature conductors per path
D. armature parallel paths
38. Load saturation characteristic of a dc generator gives relation between
- A. V and I_a C. E and I_a
B. E_o and I_f D. V and I_f
39. Under constant load conditions, the speed of a dc motor is affected by
- A. field flux
B. armature current
C. back emf
D. Both B and C
40. A wave winding must go at least _____ around the armature before it closes back where it started.
- A. once C. twice
B. thrice D. four times



41. An ideal dc generator is one that has _____ voltage regulation.

- A. low
- B. positive
- C. zero
- D. negative

42. It is possible to increase the field flux and, the same time, increase the speed of a dc motor provided its _____ is held constant

- A. applied voltage
- B. torque
- C. armature current resistance
- D. armature current

43. In a dc generator, the effect of armature reaction on the main pole flux is to

- A. reduce it
- B. distort it
- C. reverse it
- D. Both A and B

44. For the voltage build-up of a self-excited dc generator, which of the following is not an essential condition?

- A. There must be some residual flux
- B. Field winding mmf must aid the residual flux
- C. Total field circuit resistance must be less than the critical value
- D. Armature speed must be very high

45. The main function of interpoles is to minimize _____ between the brushes and the commutator when the dc machine is loaded.

- A. friction
- B. wear and tear
- C. sparking
- D. current

46. The induced emf in the armature conductors of a dc motor is

- A. sinusoidal
- B. trapezoidal
- C. rectangular
- D. alternating

47. If residual magnetism of a shunt generator is destroyed accidentally, it

may be restored by connecting its shunt field

- A. to earth
- B. to an ac source
- C. in reverse
- D. to a dc source

48. Lap winding is suitable for _____ current, _____ voltage dc generators.

- A. high, low
- B. low, high
- C. low, low
- D. high, high

49. Which of the following quantity maintains the same direction whether a machine runs as a generator or as a motor?

- A. Induced emf
- B. Armature current
- C. Field current
- D. Supply current

50. The dc armature winding in which coil sides are a pole pitch apart is called _____ winding.

- A. multiplex
- B. fractional-pitch
- C. full-pitch
- D. pole-pitch

END